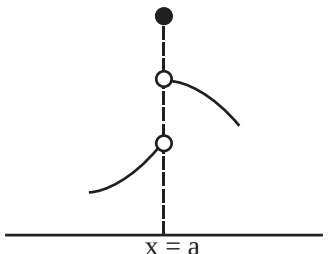
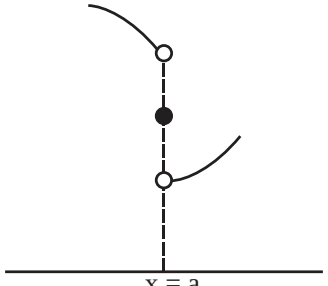
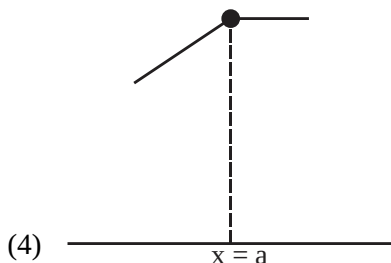
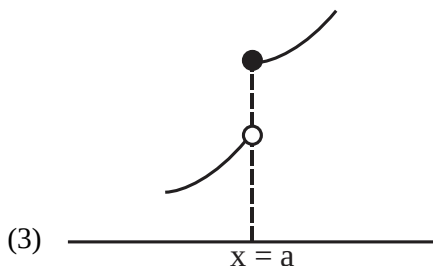


Lakshya JEE 2.0 (2024)

Application of Derivative

DPP-04

- The function $f(x) = x^3 - 3x^2 - 24x + 5$ is an increasing function in the interval given below:
 (1) $(-\infty, -2) \cup (4, \infty)$ (2) $(-2, \infty)$
 (3) $(-2, 4)$ (4) $(-\infty, 4)$
- Which one is the correct statement about the function $f(x) = \sin 2x$
 (1) $f(x)$ is increasing in $\left(0, \frac{\pi}{2}\right)$ and decreasing in $\left(\frac{\pi}{2}, \pi\right)$
 (2) $f(x)$ is decreasing in $\left(0, \frac{\pi}{2}\right)$ and increasing in $\left(\frac{\pi}{2}, \pi\right)$
 (3) $f(x)$ is increasing in $\left(0, \frac{\pi}{4}\right)$ and decreasing in $\left(\frac{\pi}{4}, \frac{\pi}{2}\right)$
 (4) The statement (1), (2) and (3) all are correct
- Let $y = x^2 e^{-x}$, then the interval in which y increases with respect to x is:
 (1) $(-\infty, \infty)$ (2) $(-2, 0)$
 (3) $(2, \infty)$ (4) $(0, 2)$
- If $f(x) = \sin x - \frac{x}{2}$ is increasing function, then
 (1) $0 < x < \frac{\pi}{3}$ (2) $-\frac{\pi}{3} < x < 0$
 (3) $-\frac{\pi}{3} < x < \frac{\pi}{3}$ (4) $x = \frac{\pi}{2}$
- The function $\sin x - \cos x$ is increasing in the interval
 (1) $\left[\frac{3\pi}{4}, \frac{7\pi}{4}\right]$ (2) $\left[0, \frac{3\pi}{4}\right]$
 (3) $\left[\frac{\pi}{4}, \frac{3\pi}{4}\right]$ (4) None of these
- Let $f(x) = 2\sin^3 x - 3\sin^2 x + 12\sin x + 5$, $0 \leq x \leq \frac{\pi}{2}$.
 Then $f(x)$, is:
 (1) decreasing on $[0, \pi/2]$
 (2) increasing on $[0, \pi/2]$
 (3) increasing on $[0, \pi/4]$ and decreasing on $[\pi/4, \pi/2]$
 (4) None of these
- If $f(x) = \frac{1}{x+1} - \log(1+x)$, $x > 0$, then f is:
 (1) A increasing function
 (2) A decreasing function
 (3) Both increasing and decreasing function
 (4) None of these
- If the function $f(x) = 3\cos|x| - 6ax + b$ increases for all $x \in \mathbb{R}$, then the range of values of a given by
 (1) $(-1/2, \infty)$ (2) $(-\infty, -1/2)$
 (3) $(-\infty, -2)$ (4) $(-2, \infty)$
- For each of the following graph, comment whether $f(x)$ is increasing or decreasing or neither increasing nor decreasing at $x = a$.

 (1) 
 (2)



10. A function is matched below against an interval where it is supposed to be increasing. Which of the following parts is incorrectly matched?

Interval	Function
(1) $[2, \infty)$	$2x^3 - 3x^2 - 12x + 6$
(2) $(-\infty, \infty)$	$x^3 - 3x^2 + 3x + 3$
(3) $(-\infty, -4]$	$x^3 + 6x^2 + 6$
(4) $\left(-\infty, \frac{1}{3}\right]$	$3x^2 - 2x + 1$

11. If $f(x) = kx^3 - 9x^2 + 9x + 3$ is monotonically increasing in $\mathbb{R}$, then

- (1) $k < 3$ (2) $k \leq 2$
(3) $k \geq 3$ (4) none of these

12. Let $g(x) = 2f\left(\frac{x}{2}\right) + f(2-x)$ and $f'(x) < 0 \quad \forall x \in (0, 2)$. If $g(x)$ increase in (a, b) and decreases in (c, d) then the value of $a+b+c+d - \frac{2}{3}$ is

13. The set of values of a for which the function $f(x) = \left(\frac{a^2-1}{3}\right)x^3 + (a-1)x^2 + 2x + 1$ increasing on $\mathbb{R}$, is

- (1) $(-3, 1)$ (2) $\mathbb{R} - [-3, 1]$
(3) $(-\infty, -3]$ (4) $[1, \infty)$

14. The values of ' a ' for which the function $(a+2)x^3 - 3ax^2 + 9ax - 1$ decreases monotonically throughout for all real x , are

- (1) $a < -2$ (2) $a > -2$
(3) $-3 < a < 0$ (4) $-\infty < a \leq -3$

Note: Kindly find the Video Solution of DPPs Questions in the DPPs Section.

Answer Key

- | | |
|--------|---------|
| 1. (1) | 10. (4) |
| 2. (3) | 11. (3) |
| 3. (4) | 12. (4) |
| 4. (3) | 13. (2) |
| 5. (2) | 14. (4) |
| 6. (2) | |
| 7. (2) | |
| 8. (2) | |
| 9. (1) | |

Neither monotonically increasing nor decreasing

(2) Monotonically decreasing

(3) Monotonically increasing

(4) Neither monotonically increasing nor decreasing



PW Web/App - <https://smart.link/7wwosivoicgd4>

Library- <https://smart.link/sdfez8ejd80if>